WORKED SOLUTION

STEP III 2010, Question 8

Pure • Quotient Rule Integration

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q8

The key idea

The expression $\mathrm{QR}' - \mathrm{Q}'\mathrm{R}$ is the numerator you get when you differentiate the quotient R/Q. So any integrand of the form P/Q^2 can be integrated at sight, provided we can find an R that produces the given P. In both parts the denominator tells us Q, and we find R by comparing coefficients. In part (ii) the left-hand side of the differential equation turns out to be exactly $\mathrm{Q}y' - \mathrm{Q}'y$, which is the same structure in disguise.

Solution

The general integral

By the quotient rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{R}(x)}{\mathrm{Q}(x)}\right) = \frac{\mathrm{Q}(x)\mathrm{R}'(x) - \mathrm{Q}'(x)\mathrm{R}(x)}{(\mathrm{Q}(x))^2} = \frac{\mathrm{P}(x)}{(\mathrm{Q}(x))^2},$$

so

$$\int \frac{\mathrm{P}(x)}{(\mathrm{Q}(x))^2}\,\mathrm{d}x = \frac{\mathrm{R}(x)}{\mathrm{Q}(x)} + C.$$

Part (i)

Here $\mathrm{Q}(x) = 1 + 2x + 3x^2$, so $\mathrm{Q}'(x) = 2 + 6x$. With $\mathrm{R}(x) = a + bx + cx^2$ and $\mathrm{R}'(x) = b + 2cx$,

$$\begin{aligned}\mathrm{QR}' - \mathrm{Q}'\mathrm{R} &= (1 + 2x + 3x^2)(b + 2cx) - (2 + 6x)(a + bx + cx^2)\\ &= (b - 2a) + (2c - 6a)x + (2c - 3b)x^2.\end{aligned}$$

(The x^3 terms, $6cx^3$ from each product, cancel.) We need this to equal $5x^2 - 4x - 3$, so

$$b - 2a = -3, \qquad 2c - 6a = -4, \qquad 2c - 3b = 5.$$

The first two give $b = 2a - 3$ and $c = 3a - 2$. Substituting into the third,

$$2(3a-2) - 3(2a-3) = 6a - 4 - 6a + 9 = 5,$$

which holds *whatever* a is. So the third equation adds no new information, and every value of a works:

$$\mathrm{R}(x) = a + (2a-3)x + (3a-2)x^2.$$

Taking $a = 1$ gives the tidiest choice, $\mathrm{R}(x) = 1 - x + x^2$, and so

$$\int \frac{5x^2 - 4x - 3}{(1 + 2x + 3x^2)^2}\,\mathrm{d}x = \frac{1 - x + x^2}{1 + 2x + 3x^2} + C.$$

The denominator never vanishes (its discriminant is $4 - 12 < 0$), so this holds for all real x.

Non-uniqueness. We have just seen that there is a one-parameter family of valid R, so the choice is not unique: for instance $a = 0$ gives $\mathrm{R} = -3x - 2x^2$ and $a = 1$ gives $\mathrm{R} = 1 - x + x^2$, both of which work. Comparing the choice for a general a with the one for $a = 0$,

$$(a + (2a-3)x + (3a-2)x^2) - (-3x - 2x^2) = a + 2ax + 3ax^2 = a\,\mathrm{Q}(x).$$

? **Ask yourself** Why should adding a multiple of Q to R make no difference?

If we replace R by $\mathrm{R} + k\mathrm{Q}$ for a constant k, then

$$\mathrm{Q}(\mathrm{R} + k\mathrm{Q})' - \mathrm{Q}'(\mathrm{R} + k\mathrm{Q}) = \mathrm{QR}' - \mathrm{Q}'\mathrm{R} + k(\mathrm{QQ}' - \mathrm{Q}'\mathrm{Q}) = \mathrm{QR}' - \mathrm{Q}'\mathrm{R},$$

so P is unchanged. Correspondingly, the answer R/Q changes to $\mathrm{R}/\mathrm{Q} + k$: the different choices of R give antiderivatives that differ only by a constant, which is absorbed into the arbitrary constant of integration. For example,

$$\frac{1 - x + x^2}{1 + 2x + 3x^2} = \frac{-3x - 2x^2}{1 + 2x + 3x^2} + 1.$$

Part (ii)

Let $\mathrm{Q}(x) = 1 + \cos x + 2\sin x$. Then $\mathrm{Q}'(x) = -\sin x + 2\cos x$, so the coefficient of y in the equation, $\sin x - 2\cos x$, is exactly $-\mathrm{Q}'(x)$. The equation is therefore

$$\mathrm{Q}\frac{\mathrm{d}y}{\mathrm{d}x} - \mathrm{Q}'y = 5 - 3\cos x + 4\sin x.$$

On any interval where $\mathrm{Q} \neq 0$ we divide by Q^2 and recognise the left-hand side as a derivative of a quotient:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\mathrm{Q}}\right) = \frac{5 - 3\cos x + 4\sin x}{\mathrm{Q}^2}.$$

(In other words, $1/\mathrm{Q}^2$ is an integrating factor.) To integrate the right-hand side we look for an R with $\mathrm{QR}' - \mathrm{Q}'\mathrm{R} = 5 - 3\cos x + 4\sin x$. Mirroring the form of Q, try $\mathrm{R} = a + b\cos x + c\sin x$, so $\mathrm{R}' = -b\sin x + c\cos x$. Expanding,

$$\mathrm{QR}' = -b\sin x + c\cos x + (2c - b)\sin x\cos x + c\cos^2 x - 2b\sin^2 x,$$
$$\mathrm{Q}'\mathrm{R} = -a\sin x + 2a\cos x + (2c - b)\sin x\cos x + 2b\cos^2 x - c\sin^2 x.$$

Subtracting, the $\sin x\cos x$ terms cancel, and the squared terms combine as $(c - 2b)(\cos^2 x + \sin^2 x) = c - 2b$. So

$$\mathrm{QR}' - \mathrm{Q}'\mathrm{R} = (c - 2b) + (c - 2a)\cos x + (a - b)\sin x.$$

Matching with $5 - 3\cos x + 4\sin x$:

$$c - 2b = 5, \qquad c - 2a = -3, \qquad a - b = 4.$$

Subtracting the second equation from the first gives $2a - 2b = 8$, the third equation again, so as in part (i) there is a free parameter, for the same reason. Taking $b = 0$ gives $c = 5$, $a = 4$, so $\mathrm{R} = 4 + 5\sin x$. Hence

$$\frac{y}{\mathrm{Q}} = \frac{4 + 5\sin x}{1 + \cos x + 2\sin x} + C,$$

and the general solution is

$$y = 4 + 5\sin x + C(1 + \cos x + 2\sin x),$$

where C is an arbitrary constant. Although we divided by Q to find it, this function satisfies the equation for every x, as direct substitution confirms. Other choices of b just shift C: for example $b = -2$ gives $\mathrm{R} = 2 - 2\cos x + \sin x = (4 + 5\sin x) - 2\mathrm{Q}$.

Checks

- In part (i), $\int_0^1$ of the integrand should be $\frac{1}{6} - 1 = -\frac{5}{6}$ from our antiderivative; Simpson's rule gives $-0.83333\ldots$, and agreement holds over random intervals for every choice of a.
- In part (ii), substitute $y = 4 + 5\sin x$: the left-hand side is $(1 + \cos x + 2\sin x)5\cos x - (2\cos x - \sin x)(4 + 5\sin x) = 5 - 3\cos x + 4\sin x$. And $y = \mathrm{Q}$ gives $\mathrm{QQ}' - \mathrm{Q}'\mathrm{Q} = 0$, the complementary function.
- Solving the equation numerically (Runge–Kutta) from $y(0.5) = 1.7$ to $x = 1.5$ gives 3.910638, matching the formula with the corresponding C.

Common slips

- Expecting three equations in three unknowns to have a unique solution. Here they are dependent, and that is the whole point of the non-uniqueness question.

- Getting the sign of Q' wrong in part (ii): $\mathrm{Q}' = 2\cos x - \sin x$, so the y coefficient is $-\mathrm{Q}'$, not $+\mathrm{Q}'$.
- Forgetting that the constant of integration multiplies Q in the final answer: the general solution is $y = \mathrm{R} + C\mathrm{Q}$, not $y = \mathrm{R} + C$.

Beyond STEP

The equation in part (ii) is a first-order linear differential equation, and the general solution has the shape "particular solution plus multiple of the solution of the homogeneous equation", a pattern that runs through all linear problems. The expression $\mathrm{QR}' - \mathrm{Q}'\mathrm{R}$ is the Wronskian of Q and R, which plays a central role in second-order linear equations, notably in reduction of order and variation of parameters. Boyce and DiPrima, *Elementary Differential Equations*, covers both integrating factors and Wronskians.

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